Question 8		
(a)	Uranium, barium, and krypton are located towards the **right-hand side of the peak** of the binding energy curve shown in Figure 8.1. Barium and krypton have a **higher binding energy per nucleon** than uranium, as uranium has a higher nucleon number Consequently, the **total binding energy** of the resulting barium and krypton nuclei is greater than the total binding energy of the initial uranium nucleus. This means that **more energy is released** when the barium and krypton nuclei are formed from free nucleons than the **energy required to separate** the uranium nucleus into its constituent nucleons. Alternative: Consequently, the **total binding energy** of the resulting barium and krypton nuclei is greater than the binding energy of the initial uranium nucleus. This means that (the total mass defect decreases after the reaction and) the **products have smaller total mass**. Hence, this nuclear fission process results in a net release of energy.	**B1** **B1**
(b)(i)	[92 ×mass of proton + 143 × mass of neutron – mass of uranium nucleus]c^2 = BE of uranium $(92\times1.00814\text{u}+143\times100898\text{u}-\text{mass}_{\text{Uranium}})c^2=235\times7.5909\times1.6\times10^{-13}$ $92\times1.00814\text{u}+143\times1.00898\text{u}-\text{mass}_{\text{Uranuim}}=1.91043\text{u}$ $\text{mass}_{\text{uranium}}$ = 235.12259u	**M1** **M1** **A0**
(b)(ii)	Energy released = (mass of reactants – mass of products)c^2 $=(235.12259-140.91441-91.92615-2\times1.00898)\text{u}c^2$ $=3.95\times10^{-11}$ J	**C1** **A1**
(iii)	The initial colliding neutron has kinetic energy hence by conservation of mass energy, the total kinetic energy of the products will need to include the kinetic energy of the colliding neutron.	**B1**